

Python List Practice — Basic to Hard

50 practice questions with Example Input and Expected Output

Level	Questions
Beginner	10
Beginner → Intermediate	10
Intermediate	10
Advanced	10
Hard	10

Beginner

Q No.	Question	Example Input	Example Output
1	Create a list of 5 integers and print the list.	[10, 20, 30, 40, 50]	[10, 20, 30, 40, 50]
2	Find the length of a list.	[4, 8, 12, 16]	4
3	Print the first element of a list.	[10, 20, 30]	10
4	Print the last element of a list.	[10, 20, 30]	30
5	Access the third element using indexing.	['a', 'b', 'c', 'd']	c
6	Print a list in reverse order using slicing.	[1, 2, 3, 4, 5]	[5, 4, 3, 2, 1]
7	Add an element to the end of a list.	[1, 2, 3], add 4	[1, 2, 3, 4]
8	Insert an element at index 1.	[10, 30], insert 20 at index 1	[10, 20, 30]
9	Remove a given element from a list.	[10, 20, 30], remove 20	[10, 30]
10	Count how many times an element occurs.	[2, 4, 2, 6, 2], count 2	3

Beginner → Intermediate

Q No.	Question	Example Input	Example Output
11	Find the maximum element in a list.	[12, 5, 27, 9]	27
12	Find the minimum element in a list.	[12, 5, 27, 9]	5
13	Find the sum of all list elements.	[10, 20, 30]	60
14	Calculate the average of list elements.	[10, 20, 30]	20.0
15	Check whether a value exists in a list.	[3, 7, 9, 12], search 9	True
16	Sort a list in ascending order.	[5, 1, 4, 2, 3]	[1, 2, 3, 4, 5]
17	Sort a list in descending order.	[5, 1, 4, 2, 3]	[5, 4, 3, 2, 1]
18	Copy a list and modify the copy.	[1, 2, 3], add 4 to copy	Original: [1, 2, 3], Copy: [1, 2, 3, 4]
19	Join two lists.	[1, 2, 3] and [4, 5, 6]	[1, 2, 3, 4, 5, 6]
20	Find the second largest element.	[10, 40, 20, 50, 30]	40

Intermediate

Q No.	Question	Example Input	Example Output
21	Remove duplicate elements from a list while preserving order.	[1, 2, 2, 3, 1, 4]	[1, 2, 3, 4]
22	Create a list containing squares of numbers from 1 to 5.	[1, 2, 3, 4, 5]	[1, 4, 9, 16, 25]
23	Create a list of even numbers from a given list.	[1, 2, 3, 4, 5, 6]	[2, 4, 6]
24	Create a list of odd numbers from a given list.	[1, 2, 3, 4, 5, 6]	[1, 3, 5]
25	Separate positive and negative numbers.	[-3, 5, -1, 8, -2, 4]	Positive: [5, 8, 4], Negative: [-3, -1, -2]
26	Find the frequency of each element.	[1, 2, 2, 3, 1, 2]	{1: 2, 2: 3, 3: 1}
27	Reverse a list without using reverse().	[1, 2, 3, 4]	[4, 3, 2, 1]
28	Find common elements between two lists.	[1, 2, 3, 4] and [3, 4, 5, 6]	[3, 4]
29	Find elements present in the first list but not the second.	[1, 2, 3, 4] and [3, 4, 5]	[1, 2]
30	Move all zeros to the end of a list.	[0, 1, 0, 3, 12]	[1, 3, 12, 0, 0]

Advanced

Q No.	Question	Example Input	Example Output
31	Find the third largest unique element.	[10, 20, 40, 40, 30, 50]	30
32	Rotate a list to the right by 2 positions.	[1, 2, 3, 4, 5]	[4, 5, 1, 2, 3]
33	Rotate a list to the left by 2 positions.	[1, 2, 3, 4, 5]	[3, 4, 5, 1, 2]
34	Find all pairs whose sum equals a target.	[2, 4, 3, 5, 7], target 7	[(2, 5), (4, 3)]
35	Flatten a one-level nested list.	[[1, 2], [3, 4], [5]]	[1, 2, 3, 4, 5]
36	Merge two sorted lists into one sorted list.	[1, 3, 5] and [2, 4, 6]	[1, 2, 3, 4, 5, 6]
37	Find the intersection of three lists.	[1, 2, 3, 4], [2, 3, 5], [0, 2, 3]	[2, 3]
38	Find the longest word in a list of words.	['cat', 'elephant', 'dog', 'tiger']	elephant
39	Group numbers into even and odd lists.	[1, 2, 3, 4, 5, 6]	Even: [2, 4, 6], Odd: [1, 3, 5]
40	Find the running/cumulative sum of a list.	[1, 2, 3, 4]	[1, 3, 6, 10]

Hard

Q No.	Question	Example Input	Example Output
41	Find the longest consecutive sequence in an unsorted list.	[100, 4, 200, 1, 3, 2]	[1, 2, 3, 4]
42	Find the maximum subarray sum.	[-2, 1, -3, 4, -1, 2, 1, -5, 4]	6
43	Find all triplets whose sum is zero.	[-1, 0, 1, 2, -1, -4]	[(-1, -1, 2), (-1, 0, 1)]
44	Merge overlapping intervals.	[[1,3], [2,6], [8,10], [9,12]]	[[1,6], [8,12]]
45	Find the missing number from 1 to n.	[1, 2, 4, 5, 6]	3
46	Find the duplicate number in a list containing values 1 to n.	[1, 3, 4, 2, 2]	2
47	Find the majority element occurring more than $n/2$ times.	[2, 2, 1, 1, 1, 2, 2]	2
48	Find the product of all elements except the current element.	[1, 2, 3, 4]	[24, 12, 8, 6]
49	Find the maximum profit from one buy and one sell.	[7, 1, 5, 3, 6, 4]	5
50	Find the longest increasing subsequence length.	[10, 9, 2, 5, 3, 7, 101, 18]	4